Code: BCS401 **Course: Analysis and Design of Algorithms**
Credits: 4.0 **L:T:P - 3:0:0**
SEE: 50 Marks **CIE: 50 Marks**
SEE Hours: 03 **Max. Marks: 100**

Prerequisites if any	Recurrence Relations, Data Structures
Learning objectives	• To learn the methods for analyzing algorithms and evaluating their performance. • To demonstrate the efficiency of algorithms using asymptotic notations. • To solve problems using various algorithm design methods, including brute force, • greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic • programming, backtracking, and branch and bound. • To learn the concepts of P and NP complexity classes

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Apply asymptotic notational method to analyze the performance of the algorithms in terms of time complexity.	Understand, Analyze
CO2	Demonstrate divide & conquer approaches and decrease & conquer approaches to solve computational problems.	Apply
CO3	Make use of transform & conquer, dynamic programming and greedy approaches to solve the given real world or complex computational problems.	Apply
CO4	Illustrate backtracking, branch & bound and P,NP and NP Complete problems	Apply

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	2	2	1									1		3	2
CO2	3	2	3									1		3	2
CO3	2	2	3									1		3	2
CO4	2	2	3									1		3	2

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

		No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
	Module – 1			
1	**INTRODUCTION:**			
1.1	What is an Algorithm?	1	0	0
1.2	Fundamentals of Algorithmic Problem Solving.	2	0	0
	FUNDAMENTALS OF THE ANALYSIS OF ALGORITHM EFFICIENCY:			
1.3	Analysis Framework	1	0	0
1.4	Asymptotic Notations and Basic Efficiency Classes	1	0	0
1.5	Mathematical Analysis of Non recursive Algorithms	1	0	0
1.6	Mathematical Analysis of Recursive Algorithms.	1	0	0
	BRUTE FORCE APPROACHES:			
	Selection Sort	1	0	0
1.8	Sequential Search	1	0	0
	Module – 2			
	BRUTE FORCE APPROACHES (contd..):			
2.1	Brute Force String Matching	1	0	0
2.2	Exhaustive Search (Travelling Salesman problem)	1	0	0
2.3	Exhaustive Search (Knapsack Problem).	1	0	0
	DECREASE-AND-CONQUER:			
2.4	Topological Sorting.	1	0	0
	DIVIDE AND CONQUER:			
2.5	Merge Sort	1	0	0
2.6	Quick Sort	1	0	0
2.7	Strassen's Matrix Multiplication.	2	0	0
	Module – 3			
	TRANSFORM-AND-CONQUER:			
3.1	Balanced Search Trees	2	0	0
3.2	Heaps and Heapsort.	2	0	0
	SPACE-TIME TRADEOFFS:			
3.3	Sorting by Counting: Comparison counting sort	2	0	0
3.4	Input Enhancement in String Matching: Horspool's Algorithm.	2	0	0
	Module – 4:			
	DYNAMIC PROGRAMMING:			
4.1	Three basic examples	1	0	0
4.2	The Knapsack Problem and Memory Functions	2	0	0
4.3	Warshall's and Floyd's Algorithms.	2	0	0
	THE GREEDY METHOD:			
4.4	Prim's Algorithm	1	0	0
4.5	Kruskal's Algorithm	1	0	0
4.6	Dijkstra's Algorithm	1	0	0

ESTD : 1946

	Module – 5			
	LIMITATIONS OF ALGORITHMIC POWER:			
5.1	Decision Trees	1	0	0
5.2	P, NP, and NP-Complete Problems.	3	0	0
	COPING WITH LIMITATIONS OF ALGORITHMIC POWER:			
5.3	Backtracking (n-Queens problem)	1	0	0
5.4	Backtracking (Subset-sum problem)	1	0	0
5.5	Branch-and-Bound (Travelling Salesman Problem),	1	0	0
	Total No. of Lecture Hours	**40**	-	-
	Total No. of Tutorial Hours		**00**	**-**
	Total No. of Practical Hours			**00**

Textbook

1. Introduction to the Design and Analysis of Algorithms, By Anany Levitin, 3rd Edition (Indian), 2017, Pearson.

Module-1:

Chapter 1 (Sections 1.1, 1.2)
Chapter 2(Sections 2.1, 2.2, 2.3, 2.4)
Chapter 3 (Section 3.1(only Selection sort), 3.2(only sequential search)

Module-2:

Chapter 3 (Section 3.4),
Chapter 4 (Sections 4.1 ,4.2)
Chapter 5 (Section 5.1, 5.2, 5.4(only Strassen's matrix multiplication)

Module-3:

Chapter 6 (Sections 6.3, 6.4)
Chapter 7 (Sections 7.1, 7.2)

Module-4:

Chapter 8 (Sections 8.1 , 8.2, 8.4)
Chapter 9 (Sections 9.1, 9.2, 9.3)

Module-5:

Chapter11 (Section 11.2, 11.3)
Chapter 12 (Sections 12.1,12.2(only Travelling Salesman Problem)

Reference books

ESTD : 1946

1. Computer Algorithms/C++, Ellis Horowitz, SatrajSahni and Rajasekaran, 2nd Edition, 2014, Universities Press.
2. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
3. Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)

Online Resources:

- Design and Analysis of Algorithms: https://nptel.ac.in/courses/106/101/106101060/

ESTD : 1946